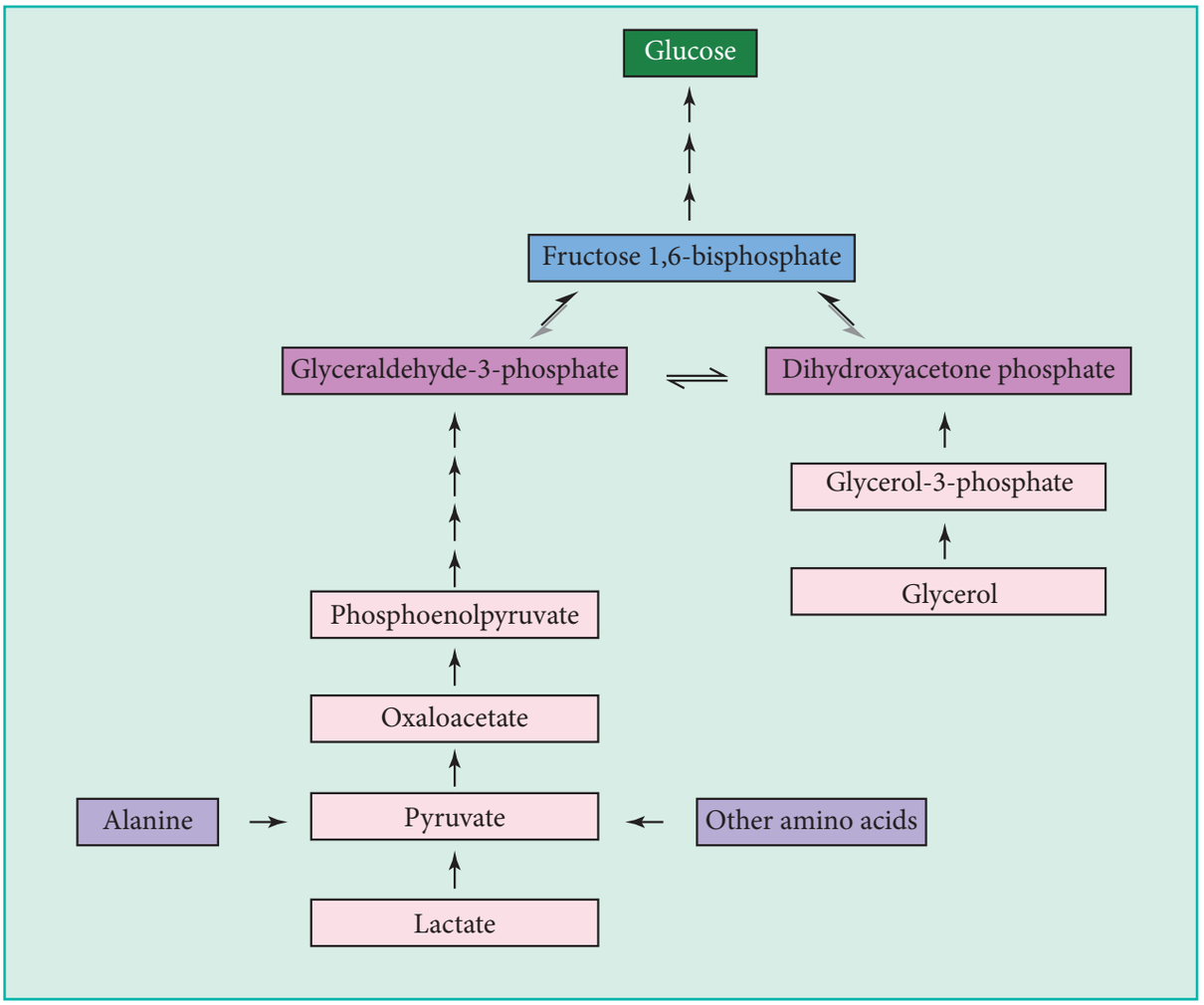


3.7.1 Key reactions in gluconeogenesis

The noncarbohydrate precursors that can be converted to glucose include the glycolysis products lactate and pyruvate, citric acid cycle intermediates, and the carbon skeletons of most amino acids. First, however, all these substances must be converted to oxaloacetate, the starting material for gluconeogenesis.

3.7.2 Reactions of gluconeogenesis

Gluconeogenesis is more or less similar to the reverse reactions of the glycolysis. However, there are some differences between these two pathways. Biological reactions can occur in both the forward and reverse direction. If the reaction occurs in the reverse direction the energetics also reversed. i.e. in the forward reaction is exergonic then the reverse reaction should be endergonic. In glycolysis, there are three irreversible steps (1, 3, 10) catalysed by the enzymes hexokinase, phosphofructokinase, and pyruvate kinase respectively. If gluconeogenesis were to simply occur in reverse direction then the reverse of the above mentioned three reactions which would require three different enzymes. The beauty of the nature is that, the above reactions of gluconeogenesis are catalysed by different enzymes namely glucose-6-phosphatase, fructose-1,6-bisphosphatase and PEP carboxykinase.

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